

JUNGUANG JIA

junguang.jia@columbia.edu | [Homepage](#) | [GitHub](#)

EDUCATION

- Columbia University** *Sep 2025 – Present*
M.A. in Statistics, Theory & Methods Honors Track
- University of California, San Diego** *Sep 2020 – Jul 2024*
B.S. in Mathematics (Probability & Statistics)
Provost Honors (multiple quarters)
- Fudan University** *Aug 2022 – Jun 2023*
B.S. in Statistics (Term Exchange)

ACADEMIC EXAMINATIONS

- Core Competency Examination** *May 2026*
Passed the examination for the Theory & Methods Honors Track, designed for doctoral preparation.

RESEARCH EXPERIENCE

- Bayesian Modeling of Pulmonary Fibrosis Trajectories** *May 2026 – Sep 2026*
Bayesian Statistics Project; Instructor: Dobrin Marchev

- Fit pooled, hierarchical, and covariate-adjusted Bayesian models in R/Stan to pulmonary fibrosis trajectories from 176 patients (1,549 measurements of forced vital capacity).
- Evaluated new-patient prediction with five-fold patient-held-out validation, prior sensitivity, and posterior predictive checks; identified residual serial dependence as a model limitation. [\[code\]](#)

- Inference for Weak OU Mean Reversion: A Reproducible Finite-Horizon Study** *Spring 2026 – Present*
Stochastic Processes Project; Instructor: Steven Campbell

- Derived and implemented exact Ornstein–Uhlenbeck likelihood and analytic nuisance profiling in Julia for irregular observation times, including the Brownian-with-drift boundary.
- Studied finite-horizon inference through information bounds and 1,600 simulated trajectories; assessed conservative split-profile inference and limited power, with independent Python/R numerical checks. [\[code\]](#)

- Stochastic Trace Estimation with Hutchinson and Hutch++: A Reproducible Computational Study** *Spring 2026 – Present*
Computational Statistics Project; Instructor: Christopher R. Harshaw

- Reconstructed unbiasedness and complexity arguments for Hutchinson and Hutch++; implemented reproducible Julia comparisons on a fixed PSD matrix and a synthetic graph.
- Compared error, matrix-vector query cost, and runtime over 1,000 trials per budget. On the fixed PSD instance, Hutch++ attained below 1% mean absolute relative error at 30 queries. [\[code\]](#)

- Volatility-Estimation Lag and Calibration Memory: A Sequential Prediction-Interval Feasibility Study** *Sep 2026 – Present*
Time Series Analysis Project; Instructor: Franz Rembart

- Analyzed volatility-estimation lag and calibration memory in rolling prediction intervals; implemented a causal Python simulation that preserves forecasts issued before each observation.
- Examined 200 synthetic scale-jump paths: EWMA normalization raised coverage over the first 25 forecasts after an upward shift from 53.7% to 83.4%, still below the 95% target in this exploratory design. [\[code\]](#)

PROJECT EXPERIENCE

ArtVenn – Cultural Heritage Data & Community Platform

Oct 2024 – Present

Founder & Technical Project Lead

- Documented sites across 200+ cities in China, recording geospatial coordinates and surviving textual and symbolic traces with provenance metadata.
- Built catalog, data-import, editorial, and search workflows with PostgreSQL, TypeScript/Next.js, Node.js, and OpenAPI; organizing the corpus for statistical analysis. [\[code\]](#)

MATHEMATICS & STATISTICS COURSEWORK

Ph.D.-Level Coursework

Computational Statistics (STAT GR6104). In progress: Theoretical Statistics I (STAT GR6201); Probabilistic Models and Machine Learning (STCS GR6701); Optimization I (IEOR E6613).

Other Selected Coursework

Honors Probability Theory

Honors Statistical Inference

Honors Linear Regression Models

Bayesian Statistics

Stochastic Processes and Applications I

Time Series Analysis

Introduction to Analysis II

Unsupervised Learning (in progress)

Theory of Diffusion Models (in progress)

Advanced Data Analysis (in progress)

WORK EXPERIENCE

Mobalytics

Jul 2023 – Sep 2023

Data Analyst, California. Analyzed player behavior across three games with SQL and regression; built Tableau dashboards and segmentation analyses for engagement and marketing recommendations.

Shanghai Dingbang International Trading Co., Ltd.

Sep 2022 – Dec 2022

Financial Assistant, Shanghai. Maintained production records and daily variance reports; coordinated reporting schedules and period-end workflows.

Hua An Securities

Jul 2022 – Sep 2022

Investment Research Analyst, Shanghai. Researched the catalyst industry and applied multiple regression and principal component analysis to support investment recommendations.

Shanghai Yunyi Huan Yu Technology Co., Ltd.

Jun 2021 – Aug 2021

R&D Assistant, Shanghai. Supported Python/Spark data integration, model analysis, and validation for O2O and SaaS platforms.

TECHNICAL SKILLS

Programming

R, Python, Julia, SQL, TypeScript, Swift

Scientific Computing

Stan/RStan, NumPy, SciPy, R Markdown

Data & Visualization

PostgreSQL, Apache Spark, Tableau

Development

Next.js/React, Node.js, OpenAPI, Docker, Linux, Git, GitHub Actions, L^AT_EX